

Coding & Solution

Date	07/07/2026
Team ID	SWTID-2026-5714
Project Name	OptiCrop: Smart Agriculture Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary



Code Quality Checklist

Field	Details
Repository / GitHub Link	https://github.com/Kushal-Antarvedi/OptiCrop
Programming Language(s)	Python
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy
Key Features Implemented	Crop prediction using Random Forest machine learning, soil & climate input form, multi-page web interface (Home, About, Predict Crop, Result), responsive UI, custom logo and favicon.
Pending / Incomplete Features	None
Setup / Run Instructions	<pre>pip install -r requirements.txt python app.py</pre> Open http://localhost:5000 in your browser